PH4101: Condensed Matter Physics

Assignment 1

`Due date: On or before Midsemester Exam`

1. Density of states for general dispersion

Consider a system in d spatial dimensions with elementary excitations obeying the dispersion relation

$$E(\mathbf{k}) = \Delta + A|\mathbf{k}|^s, \tag{1}$$

where $A > 0$, $s > 0$, and $\Delta \geq 0$.

(a) Determine the number of states with energy less than E for a system of volume $\mathcal{V}$.

(b) Show that the density of states is

$$g(E) = \frac{\mathcal{V} S_{d-1}}{s(2\pi)^d A^{d/s}} (E - \Delta)^{\frac{d}{s}-1}, \qquad E \geq \Delta, \tag{2}$$

where

$$S_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)} \tag{3}$$

is the surface area of the unit sphere in d dimensions.

(c) Using the above expressions, obtain the energy dependence of the density of states for:

(i) a one-dimensional quadratic dispersion;

(ii) a two-dimensional quadratic dispersion; and

(iii) a one-dimensional linear dispersion.

(d) Determine the conditions on d and s under which the density of states vanishes, remains finite, or diverges as $E \to \Delta^+$.

(e) Consider a one dimensional case with linear dispersion, that is, $s = 1$ and $\Delta \geq 0$. Determine the Debye cutoff k_D by imposing the normalization condition

$$\int_{\Delta}^{E_D} g_D(E)\, dE = N, \tag{4}$$

where $E_D = \Delta + Ak_D$. Write the DOS in terms of E_D.

(f) Using the Bose distribution, $n(E) = \frac{1}{e^{\beta E}-1}$ where $\beta = \frac{1}{k_B T}$, find the average internal energy using,

$$U(T) = \int_{\Delta}^{E_D} \mathrm{d}E\ E g(E) n(E) \tag{5}$$

(g) Obtain the specific heat using $C_v = \frac{\partial U}{\partial T}$ and analyse the low temperature behaviour (with respect to appropriate energy scales) for (i) $\Delta \neq 0$ (gapped) and (i) $\Delta = 0$ (gapless). Comment on the differences and their physical implication.

(h) Obtain the high temperature limit in both cases. State whether the result depend on whether the dispersion is gapped or gapless?

Useful Identity:

$$\int_0^\infty \frac{x}{e^x - 1}\, dx = \frac{\pi^2}{6}. \tag{6}$$

2. Vibrations of a one-dimensional chain with competing interactions

Consider a one-dimensional chain of N identical atoms, each of mass M, separated by an equilibrium lattice spacing a. The atoms are coupled by nearest-neighbour springs of strength $K_1 > 0$ and next-nearest-neighbour springs of strength K_2. The potential energy of the chain is

$$V = \frac{K_1}{2} \sum_{n=1}^{N} (u_{n+1} - u_n)^2 + \frac{K_2}{2} \sum_{n=1}^{N} (u_{n+2} - u_n)^2 , \tag{7}$$

where u_n denotes the displacement of the nth atom from its equilibrium position.

(a) Write down the equations of motion for u_n.

(b) By substituting a plane-wave solution,

$$u_n(t) = u_0 e^{i(kna - \omega t)}, \tag{8}$$

show that

$$\omega^2(k) = \frac{4}{M} \left[K_1 \sin^2\left(\frac{ka}{2}\right) + K_2 \sin^2(ka) \right]. \tag{9}$$

(c) Determine the long-wavelength sound velocity,

$$v_s = \lim_{k \to 0} \frac{\omega(k)}{|k|}. \tag{10}$$

(d) Find the condition on K_2 for which the lattice remains dynamically stable.

(e) Show that the density of vibrational states can be expressed as

$$g(\omega) = \frac{L}{2\pi} \sum_{k_i} \left| \frac{d\omega}{dk} \right|_{k_i}^{-1}, \tag{11}$$

where $L = Na$, and the sum runs over all wavevectors k_i satisfying $\omega(k_i) = \omega$. Verify that

$$\int_0^{\omega_{\max}} g(\omega)\, d\omega = N. \tag{12}$$

(f) For

$$K_2 = \frac{K_1}{4}, \tag{13}$$

show that the dispersion near the Brillouin-zone boundary satisfies

$$\omega_{\max} - \omega(k) \propto \left(\frac{\pi}{a} - |k| \right)^4. \tag{14}$$

Hence determine the behaviour of $g(\omega)$ close to $\omega_{\max}$.

3. Two-dimensional tight-binding model

Consider a two-dimensional square lattice containing $N = N_x N_y$ sites, with lattice constant a and one atomic orbital per site. Electrons can hop only between nearest-neighbour sites. The tight-binding Hamiltonian is

$$H = -t \sum_{\langle i,j \rangle} \left(c_i^\dagger c_j + c_j^\dagger c_i \right), \qquad t > 0. \tag{15}$$

(a) Assuming periodic boundary conditions, introduce the Fourier transformation

$$c_i = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}_i} c_{\mathbf{k}}, \tag{16}$$

to diagonalize the Hamiltonian in k-space. Show that the energy dispersion is

$$\varepsilon(\mathbf{k}) = -2t \left[\cos(k_x a) + \cos(k_y a) \right]. \tag{17}$$

Identify the first Brillouin zone and verify that the band contains N single-particle states.

(b) Determine the minimum energy, maximum energy, and bandwidth and plot the equal energy contours in $k_x - k_y$ space in the first Brillouin Zone, that is, $k_x \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right)$ and $k_y \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right)$.

(c) Compute the density of states using the dispersion $E(\mathbf{k})$ in terms of elliptic integral function $K(m)$, defined as:

$$K(s) = \int_0^{\pi/2} \mathrm{d}\theta \, \frac{1}{\sqrt{1 - s \sin^2 \theta}} \tag{18}$$

and examine the van Hove singularities in the DOS. To determine the singularities, you can consider an alternate expression of the DOS,

$$g(\epsilon) = \frac{1}{(2\pi)^2} \int_{\mathbf{k}' \in S(\epsilon)} \frac{\mathrm{d}^2\mathbf{k}'}{|\nabla E(\mathbf{k}')|} \tag{19}$$

where $S(\epsilon)$ is the constant energy surface, that is, containing all $\mathbf{k}$ such that $E(\mathbf{k}) = \epsilon$.

4. Van der Waal's interaction

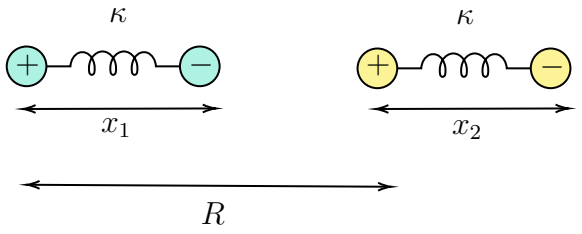
To understand the origin of the Van der Waals interaction, consider the toy model of a one-dimensional atom where the electron with charge $-e$ is attached to a fixed nucleus of charge $+e$ by a spring with spring constant κ. The atom has a dipole moment

$$\mathbf{p} = -ex\hat{\mathbf{x}} \tag{20}$$

directed towards the nucleus, where x is the displacement of the electron with respect to the nucleus, undergoing simple harmonic motion. Two such atoms are placed along a line at a distance R ($R \gg x_1, x_2$). The instantaneous displacements of the two electrons from their respective nuclei are denoted by x_1 and x_2. Assume both atoms have the same spring constant κ.

(a) Consider the system to be classical and write down the Hamiltonian of the system by considering the effective Coulomb interaction between the four charges. Show that, to leading order, the interaction energy goes as

$$E_{\text{int}} \sim R^{-3}.$$



(b) Write down the equations of motion for the negatively charged particles and comment on the form of the equation in the limit $R \to \infty$.

(c) Perform the normal mode analysis for this system and find the frequencies of oscillations ω_+ and ω_-.

(d) Considering these normal modes to be quantum mechanical harmonic oscillators, with ground state energies

$$\frac{\hbar\omega_+}{2} \qquad \text{and} \qquad \frac{\hbar\omega_-}{2},$$

show that the leading order energy correction goes as

$$E \sim R^{-6}.$$

(e) Considering the initial Hamiltonian in (a) to be quantum mechanical, treat the Coulombic interaction as a small perturbation, and find the leading order correction to the ground state energy of the coupled system.